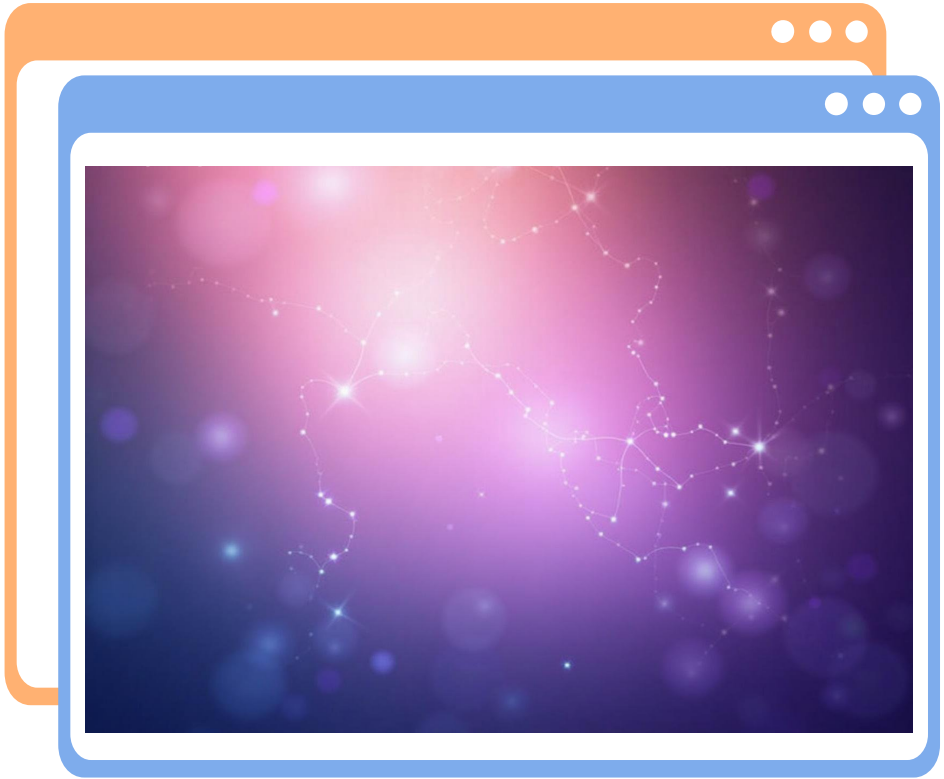
Cancer AI

Neural-network based cancer classification pipeline overview

Introduction

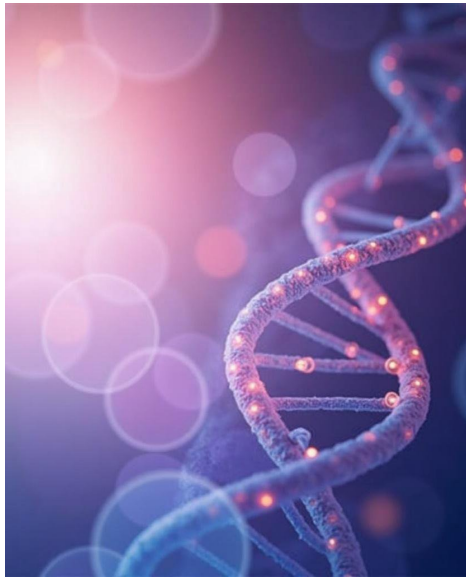
Cancer classification using neural networks aims to categorize tumor types and predict outcomes from clinical and molecular data.

Focus is on clinical-grade **accuracy** and **robustness**, combining imaging, genomics, and pathology for actionable insights.



01

Model Architecture



Data Inputs

Use *imaging* (radiology, histopathology), *genomic* profiles, and clinical records as inputs.

Apply standardized preprocessing, normalization, augmentation, and address class imbalance with sampling or weighting to ensure representative training.

Network Design

Employ *CNNs* for images and *transformers* or MLPs for sequence/tabular data; consider multi-modal fusion.

Incorporate transfer learning, cross-validation, and emphasize **explainability** for clinical trust and regulatory acceptance.

Training Strategy

Optimize with *cross-validation*, class weighting, and progressive augmentation to address imbalance and overfitting.

Use scheduled learning rates, early stopping, and ensemble models to improve generalization and **robust accuracy** across cohorts.

02

Clinical Deployment

Validation Metrics

Report *sensitivity*, *specificity*, and AUC alongside calibration and decision-curve analysis.

Include external cohort validation and prospective testing to demonstrate **clinical performance**.

Regulatory Compliance

Maintain traceable model development records, risk assessments, and explainability artifacts.

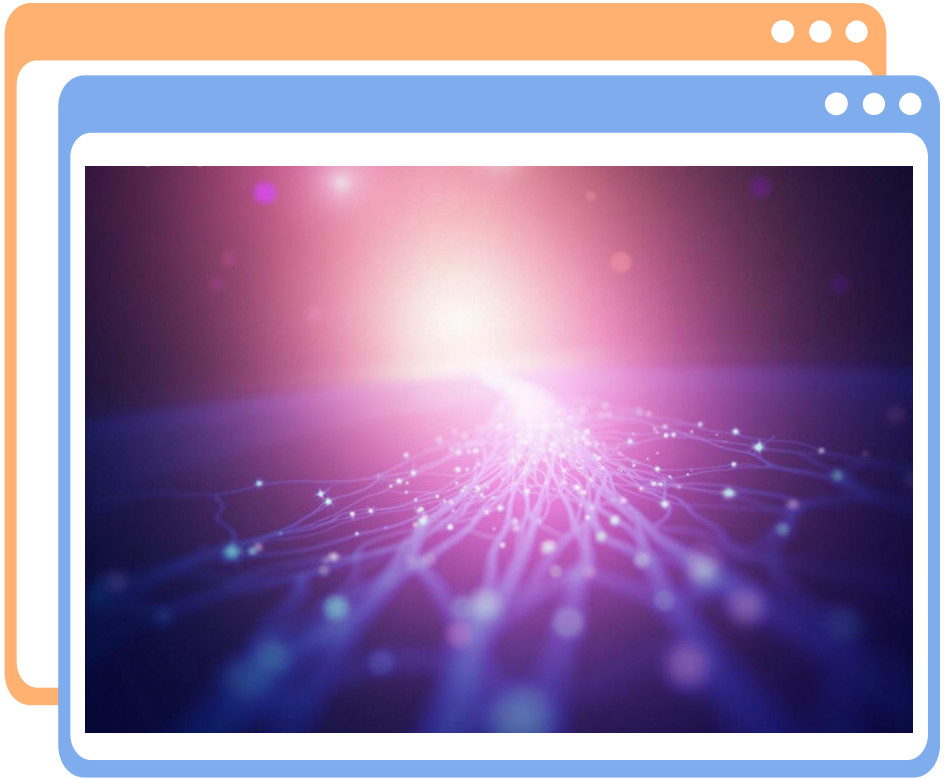
Align with data protection laws and relevant standards (e.g., FDA/CE) for medical device submissions.

Workflow Integration

Integrate with EHRs and PACS, provide clear clinician-facing outputs and alerts, and enable seamless feedback loops.

Implement monitoring dashboards and automated retraining triggers to sustain performance.

Conclusions

Combine robust training, comprehensive validation, and regulatory readiness to transition *neural models* into clinical use.

Prioritize explainability, interoperability, and continuous monitoring to ensure safe, effective deployment.